

Shashank Modi

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EDUCATION

Vellore Institute of Technology

B.Tech Computer Science; CGPA: 9.4/10

2022 – 2026

Vellore, India

D.A.V Sushil Kedia Vishwa Bharati

CBSE – 12th Grade; Percentage: 96.4%

2022

Kathmandu, Nepal

WORK EXPERIENCE

Software Development Intern

Star Union Dai-ichi Life Insurance

Jan 2026 – Present

Mumbai, Maharashtra

- Optimized a high-volume stored procedure operating on large datasets, reducing query execution time by 2x and simplifying code by 3x through indexing, join restructuring, and use of temporary tables.
- Developed a streaming XML export pipeline using .NET (XMLWriter, GZipStream, SqlDataReader), reducing memory usage to 5–20MB for large regulatory data exports.
- Migrated legacy APIs from Python to .NET, aligning services with the organization's backend ecosystem.
- Documenting legacy systems by creating system design documentation for undocumented applications.

Software Engineer Summer Intern

Providence Global Center

Jun 2025 – Aug 2025

Hyderabad, Telangana

- Developed a workflow management platform using React, ASP.NET Core, and MS SQL Server, replacing manual SharePoint-based processes across teams.
- Designed and implemented REST APIs and automated reporting workflows using MailKit, improving process visibility and reducing manual effort.
- Built real-time dashboards for facility onboarding metrics, enabling better tracking and decision-making for leadership.

Technical Intern

Biomed International

Jun 2024 – Jul 2024

Kathmandu, Nepal

- Redesigned product pages using HTML, CSS, and JavaScript, improving responsiveness and overall navigation.
- Developed a secure patient portal and resolved user-reported issues, reducing recurring frontend bugs by 15%.

PROJECTS

blip. | Full-Stack PWA(Progressive Web App) | [React.js](#), [Node.js](#), [PostgreSQL](#), [Vercel](#) [Live Demo](#)

- Engineered a rule-based NLP engine for real-time expense logging, achieving <100ms processing by eliminating LLM call latency.
- Architected a social debt settlement system using PostgreSQL ACID transactions and row-level locking for safe concurrent balance updates.
- Reduced perceived latency via optimistic UI updates and serverless warm-up strategies to mitigate cold starts.
- Designed a high-contrast, minimalist interface optimized for rapid data entry and power-user workflows.

Mental Fatigue Estimation | [Python](#), [scikit-learn](#), [XGBoost](#), [NumPy](#), [Pandas](#)

[View Code](#)

- Built an ML pipeline to estimate cognitive fatigue levels (1–5) using keystroke dynamics and mouse telemetry data.
- Engineered features such as typing latency and cursor movement patterns to convert raw input events into predictive signals.
- Evaluated Linear Regression, Random Forest, and XGBoost models; achieved best performance with XGBoost (MAE: 0.93, MSE: 0.95) outperforming linear models by capturing non-linear behavioral patterns.

SKILLS

Programming Languages: C++, Java, Python

Frameworks & Libraries: ASP.NET Core, React.js, scikit-learn, XGBoost, Material UI

Tools & Platforms: Git, Vercel, Render

Databases: MySQL, MS SQL Server, PostgreSQL (indexing, stored procedures, TVPs)

Core Concepts: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks

ACHIEVEMENTS

- Solved 250+ coding problems on LeetCode | [LeetCode Profile](#)
- Co-Secretary, Matrix Club (VIT): Conducted technical workshops and coordinated activities for 20+ members.